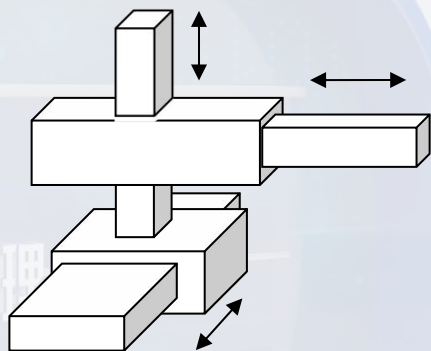


连杆速度的传递与计算



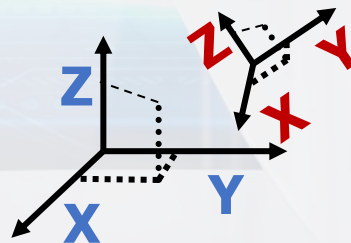
连杆速度的传递与计算

平移速度 V



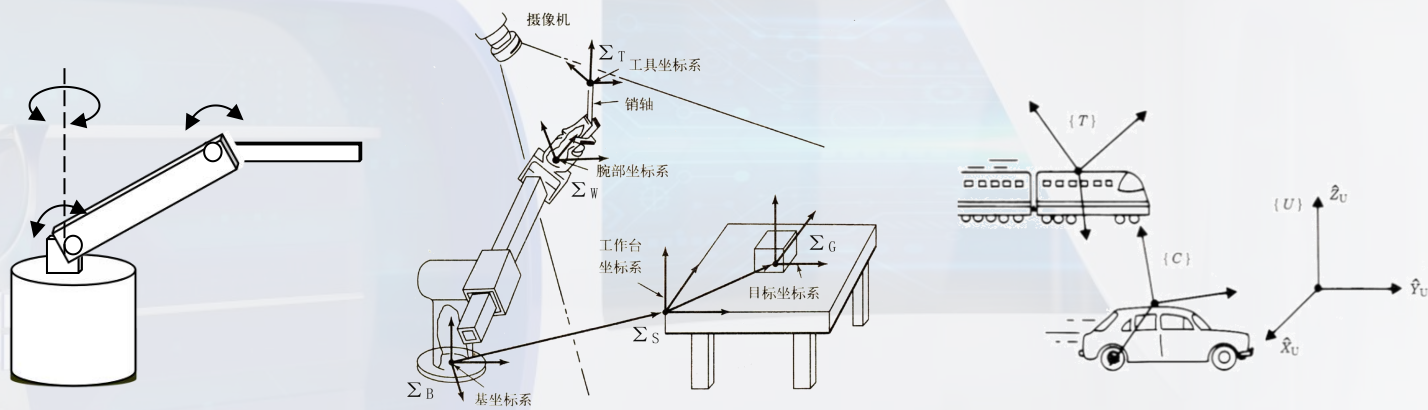
速度的相对性：

✓ 速度是在那个坐标系描述的？



连杆速度的传递与计算

角速度 ω



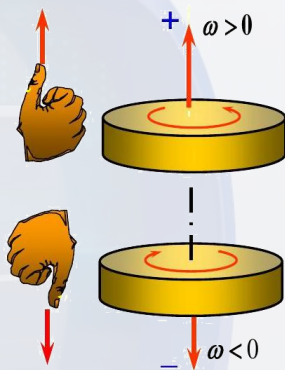
角速度的概念及特点

刚体的旋转



角速度的概念及特点

旋转轴



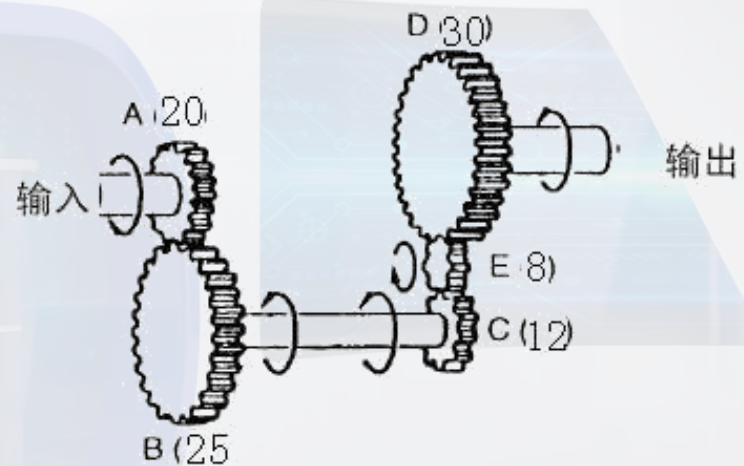
$$\omega = d\theta/dt$$

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$$



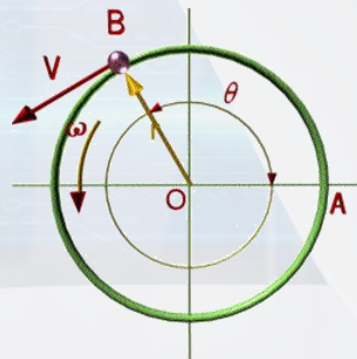
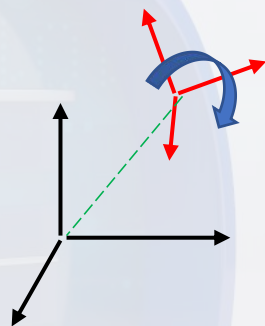
角速度的概念及特点

如图所示，请计算传动比 i_{00}



角速度的概念及特点

坐标系的旋转



相邻连杆间的速度关系

$${}^i \mathbf{V}_{i+1} = {}^i \mathbf{V}_i + {}^i \boldsymbol{\omega}_i \times {}^i \mathbf{P}_{i+1}$$

$${}^i \boldsymbol{\omega}_{i+1} = {}^i \boldsymbol{\omega}_i + {}_{i+1}^i \mathbf{R} \dot{\theta}_{i+1} {}^{i+1} \mathbf{Z}_{i+1}$$

$${}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -d_i s\alpha_{i-1} \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & d_i c\alpha_{i-1} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

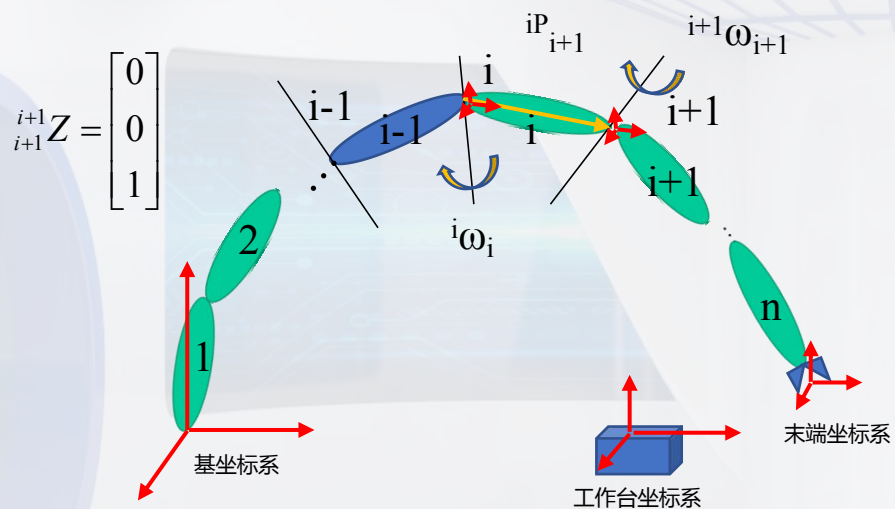
$${}^{i+1}T_i = \begin{bmatrix} c\theta_{i+1} & -s\theta_{i+1} & 0 & a_i \\ s\theta_{i+1} c\alpha_i & c\theta_{i+1} c\alpha_i & -s\alpha_i & -d_{i+1} s\alpha_i \\ s\theta_{i+1} s\alpha_i & c\theta_{i+1} s\alpha_i & c\alpha_i & d_{i+1} c\alpha_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



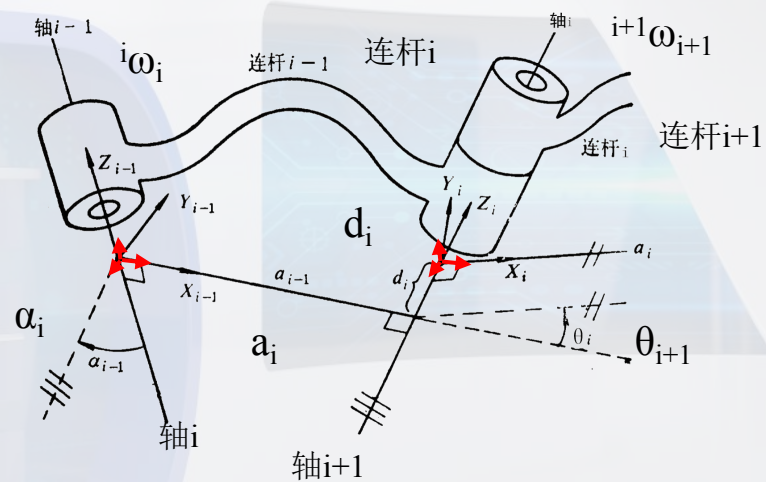
相邻连杆间的速度关系

$${}^{i+1}_i P = \begin{bmatrix} a_i \\ -d_{i+1} s \alpha_i \\ d_{i+1} c \alpha_i \end{bmatrix}$$

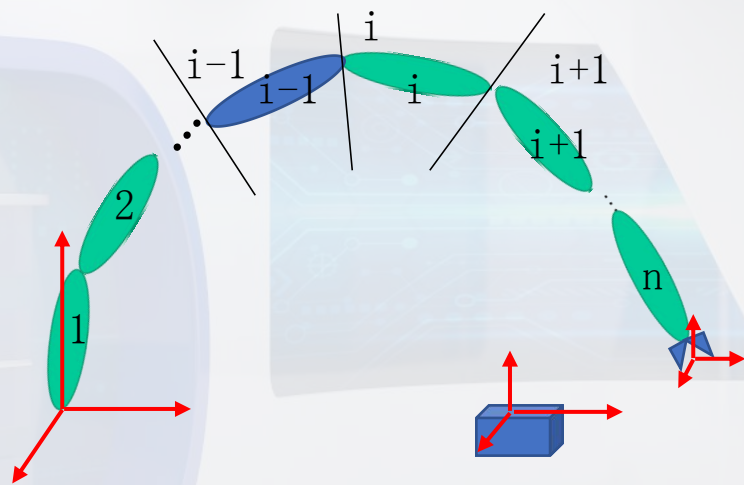
$${}^A_B T = \begin{bmatrix} {}^A_B R & {}^A P_{BO} \\ 0 & 1 \end{bmatrix}$$



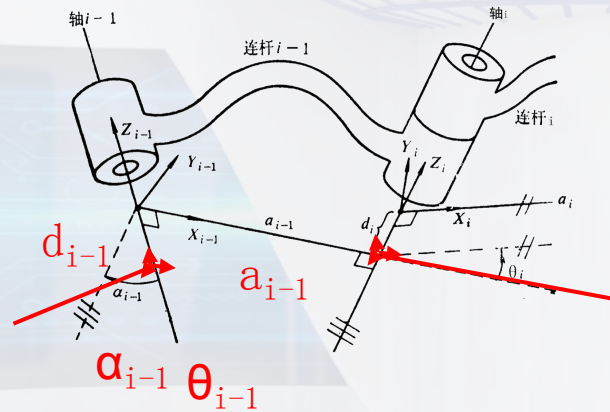
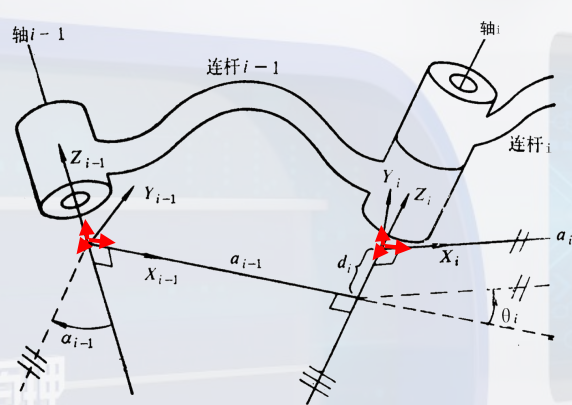
相邻连杆间的速度关系



连杆坐标系的两种建立方式



连杆坐标系的两种建立方式



在连杆n坐标系中描述速度

当连杆i+1为转动轴时：

$${}^{i+1}\boldsymbol{\omega}_{i+1} = {}^{i+1}_i\mathbf{R}^i\boldsymbol{\omega}_i + \dot{\boldsymbol{\theta}}_{i+1} {}^{i+1}\mathbf{Z}_{i+1}$$

$${}^{i+1}\mathbf{v}_{i+1} = {}^{i+1}_i\mathbf{R}\left({}^i\mathbf{v}_i + {}^i\boldsymbol{\omega}_i \times {}^i\mathbf{P}_{i+1}\right)$$

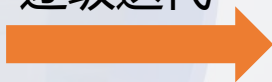
在连杆n坐标系中描述速度

当连杆i+1为移动轴时：

$${}^{i+1}\omega_{i+1} = {}^{i+1}_i R {}^i\omega$$

$${}^{i+1}v_{i+1} = {}^{i+1}_i R ({}^i v_i + {}^i\omega_i \times {}^i P_{i+1}) + \dot{d}_{i+1} {}^{i+1}Z_{i+1}$$

逐级迭代



${}^n\omega_n$

nV_n

END